

The empirical recurrence for OEIS A295414: recovering a conjecture that is not written down

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Abstract

OEIS A295414 records “Empirical recurrence of order 78”, with the recurrence itself in a linked file rather than in the entry. The conjecture can nevertheless be settled, and the recurrence recovered, without reading that file. The entry counts arrays under a condition local to a bounded window of lines, so the count is a walk count on a finite digraph and satisfies some monic recurrence of order at most the number of its states with distinct futures, here $S' = 120$; Berlekamp–Massey applied to $2S'$ exact terms therefore returns the MINIMAL such recurrence. Its order is exactly 78, the order the entry states, and the same is true of every tail of the sequence. Any recurrence of order 78 that the sequence satisfies from any point on must then have a characteristic polynomial that is a multiple of the minimal one and of the same degree, hence equal to it. So the entry’s recurrence is the one displayed here, whatever its file contains, and it is proved.

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1 A conjecture one cannot read

OEIS A295414 is “Number of nX6 0..1 arrays with each 1 horizontally or vertically adjacent to 1, 2 or 3 1s.”. It has offset 1 and begins

21, 1632, 68613, 3019079, 136204120, 6093564449, 272714953724, 12210363190599, ...

The entry states:

Empirical recurrence of order 78 (see link above)

As of the “Last modified” line on the live entry (Nov 22 2017 10:38:35, revision 4) this is still recorded as empirical. The recurrence’s coefficients are not in the entry text.

2 The count is a walk count

The entry’s condition is local: it constrains a bounded window of consecutive lines of the array and nothing further. The admissible configurations of one such window are the vertices of a finite digraph, an edge joining u to v when v may follow u , and an admissible array is exactly a walk. Hence, with M the adjacency matrix on $S = 4096$ vertices, $a(n) = \iota^\top M^{j(n)} \tau$ for a linear reindexing j fixed by the entry’s shape and checked against its published terms. States with the same future contribute identically to $\iota^\top M^n \tau$, so they may be merged without changing any count; merging leaves $S' = 120$ of the 4096, and it is that smaller bound the computation below uses. By the Cayley–Hamilton theorem M^S is an integer combination of $I, M, \dots, M^{S-1}$, so:

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 120$.*

3 Recovering the recurrence

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S is determined, together with its minimal recurrence, by its first $2S$ terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Running it on $2S' = 240$ exact terms of the model returns order 78 — the order the entry states — with the integer coefficients

$$a(n) = \sum_{i=1}^{78} c_i a(n-i),$$

c_1	33	c_{21}	−355241724761337898	c_{41}	212169608866106214564303	c_{61}	−1
c_2	195	c_{22}	−4044643810164990236	c_{42}	869472908888472358058663	c_{62}	9
c_3	12104	c_{23}	1576227415995162856	c_{43}	−39483906591467654442908454	c_{63}	−
c_4	91315	c_{24}	30913337712075143165	c_{44}	73331519148699318059562379	c_{64}	2
c_5	1283839	c_{25}	−379193490000339829	c_{45}	−210321111424696044985468671	c_{65}	−
c_6	6860268	c_{26}	−163218197458492383414	c_{46}	332584615334197791873769418	c_{66}	−
c_7	2025546	c_{27}	−40461742197911107051	c_{47}	−640244554930770086785313811	c_{67}	4
c_8	−368994850	c_{28}	414645482552702099234	c_{48}	892553479087729493051169887	c_{68}	−
c_9	−8137833375	c_{29}	111149301660782995559	c_{49}	−1385913280893643951542751564	c_{69}	3
c_{10}	−17408676417	c_{30}	1333301410360151092247	c_{50}	1735422807831040595876156064	c_{70}	−
c_{11}	485203788676	c_{31}	1447052896725811492039	c_{51}	−2321935649715691351623185636	c_{71}	−
c_{12}	1827561272199	c_{32}	−14485574160834509629440	c_{52}	2645594006322222790483232612	c_{72}	−
c_{13}	−15092543151738	c_{33}	−6967094588923566879937	c_{53}	−3148724619756536180611865728	c_{73}	−
c_{14}	−71178528345874	c_{34}	17859924135794627796619	c_{54}	3289180774209246274523544512	c_{74}	−
c_{15}	308453709010693	c_{35}	−51855688407059047768727	c_{55}	−3534322502531773960034431516	c_{75}	−
c_{16}	1735652939651345	c_{36}	231192269908656001346840	c_{56}	3384713286854656015438254160	c_{76}	−
c_{17}	−4490056937388994	c_{37}	213585475578748708674347	c_{57}	−3291025040743864070403815968	c_{77}	−
c_{18}	−30265205884253295	c_{38}	−4652109958982831891369	c_{58}	2857927688008817513283462912	c_{78}	−
c_{19}	47596555622318145	c_{39}	1862443917105262984830897	c_{59}	−2490723106360912697602437312		
c_{20}	398933455151564449	c_{40}	−4126469973874636546888887	c_{60}	1909102871239511284819545536		

Theorem 3. *The recurrence above holds for every n , and it is the recurrence the entry records.*

Proof. It holds by Lemma 2, which returns a recurrence the sequence satisfies from its first term. For the identification: the entry asserts that the sequence satisfies SOME recurrence of order 78, possibly only from some index on. Let q be that recurrence’s characteristic polynomial and $q_{\min}$ the minimal polynomial of the tail of the sequence. Then $q_{\min} \mid q$. Berlekamp–Massey run on the sequence with its first $78 + 4$ terms removed returns order 78 as well, so $\deg q_{\min} = 78 = \deg q$; both are monic, so $q = q_{\min}$, which is the polynomial computed above. $\square$

4 Verification

Three checks.

First, the walk model reproduces all 14 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters.

Second, the recovered recurrence was evaluated directly on those published terms, with no matrices and no Berlekamp–Massey involved, and holds at every index where the entry gives enough earlier terms to test it.

Third, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once in exact rational arithmetic; and once more on the sequence with its first $78 + 4$ terms removed, which is what the identification above requires. All three agree.

Remark 4. What is unusual here is that the conjecture's statement was never read. The entry pins it down to a one-parameter family — the recurrences of order 78 that this sequence satisfies — and that family turns out to have exactly one member.

References

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